

How to Measure Legislative District Compactness If You Only Know it When You See it¹

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Hubert M. Blalock Memorial Lecture, University of Michigan, 7/12/2017

¹Based on joint work with Aaron Kaufman and Mayya Komisarchik

²GaryKing.org

Redistricting Defines Democracy — & Needs Fixing

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 - Required in many other jurisdictions

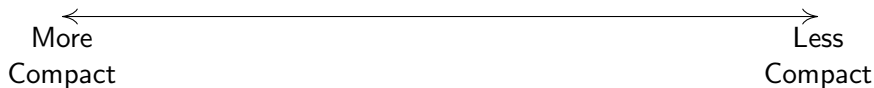
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A simple single compactness dimension that you know when you see

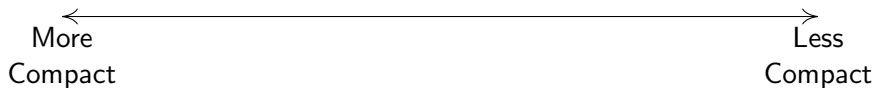
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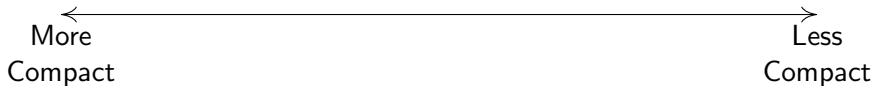
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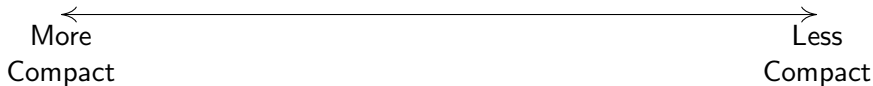
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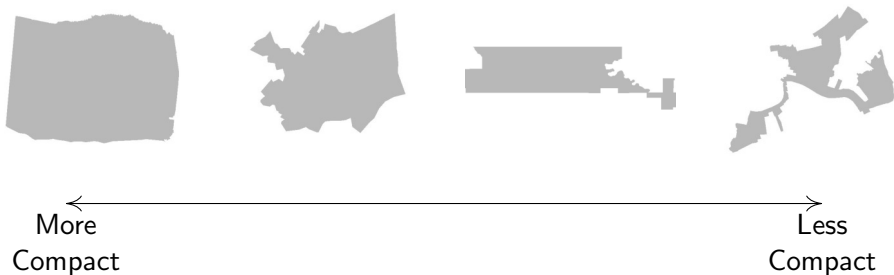
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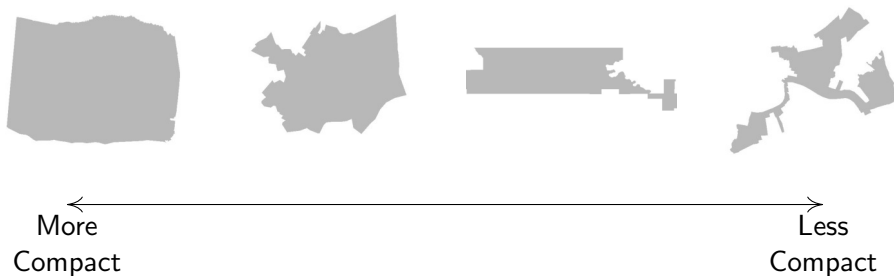
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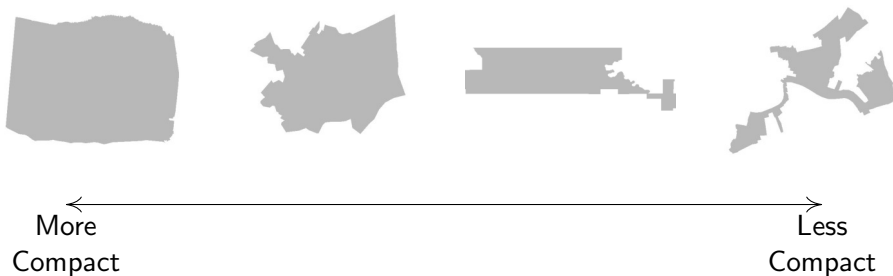
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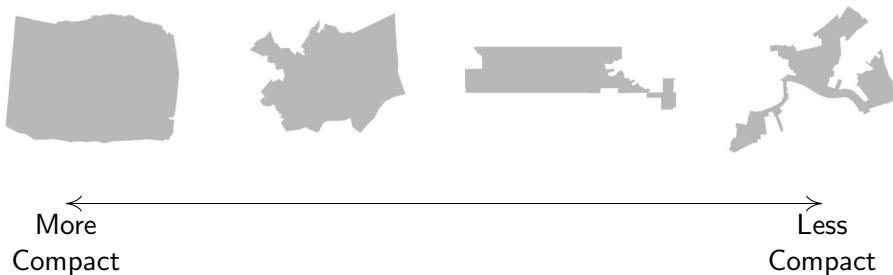
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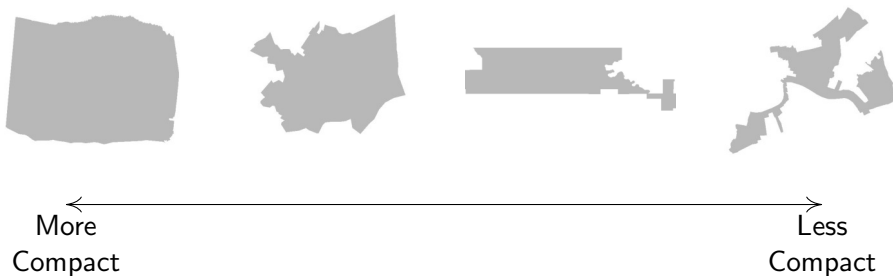
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- $\rightsquigarrow$ **Let's start with existing measures by social scientists**

Measure 1: Length/Width Ratio of Min Bounding Box

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Squarish districts more compact than long thin ones

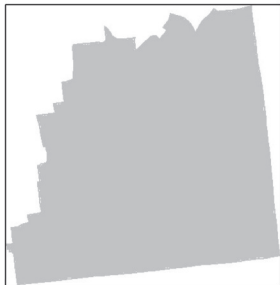
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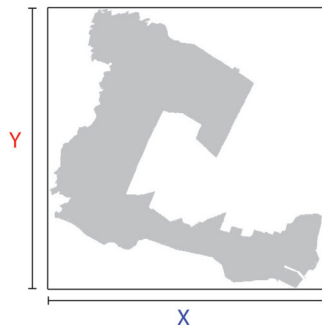
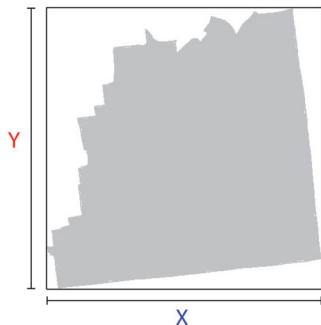
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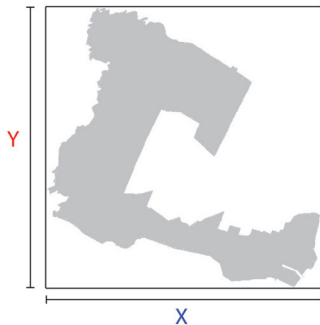
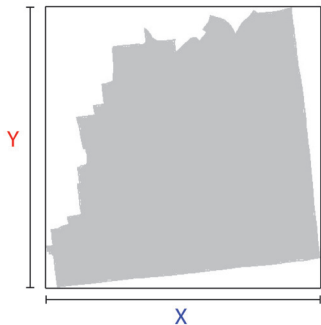
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In both districts: $X/Y \approx 1.30$

Measure 2: Reock, District / Bounding Circle Areas

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Circular districts are most compact

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Circular districts are most compact



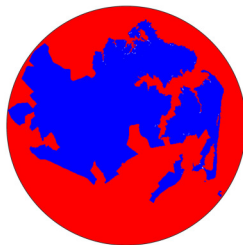
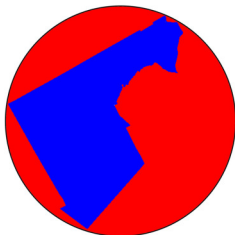
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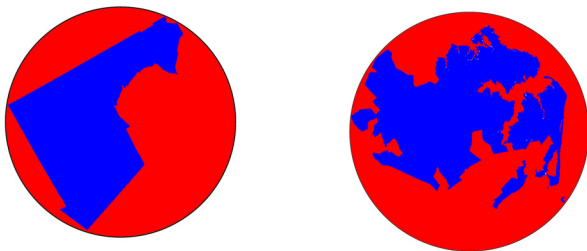
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In both cases, $X/(Y + X) \approx 0.37$

Measure 3: Boyce-Clark, Variation in Centroid Deviations

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In both cases, $\text{MAD}(r) \approx 0.31$

A Brief Rotational Invariance Interlude:

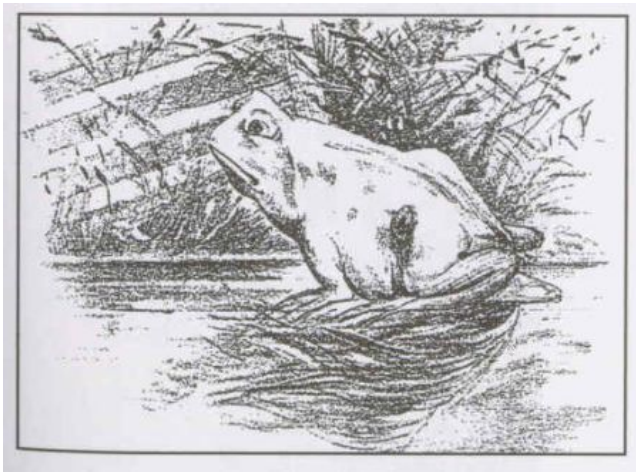
A Brief Rotational Invariance Interlude: Can you Name this Celebrity?



A Brief Rotational Invariance Interlude: Can you Name this Celebrity?



A Brief Rotational Invariance Interlude: See the Frog?



A Brief Rotational Invariance Interlude: See the Frog Horse?



Human Perception: Not Rotationally Invariant

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- ↪ Measuring “you know it when you see it”: No rotational invariance

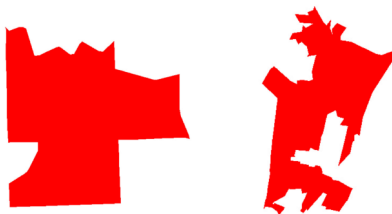
New Measure: Y-Symmetry, area of symmetric reflection

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Symmetric figures (circles, squares) are more compact

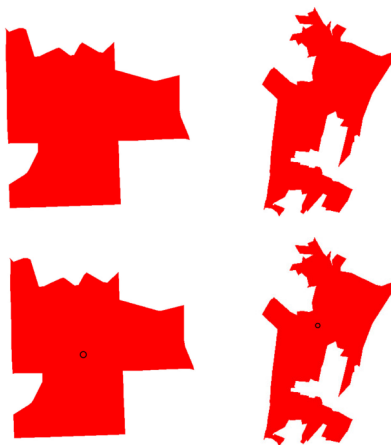
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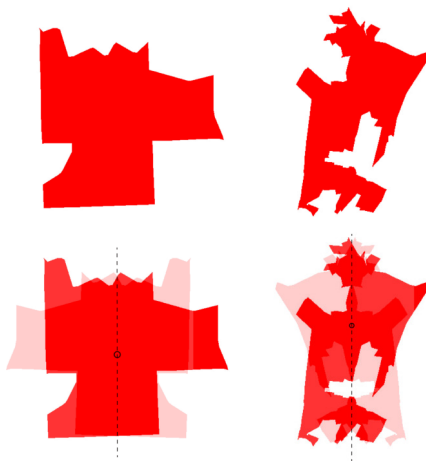
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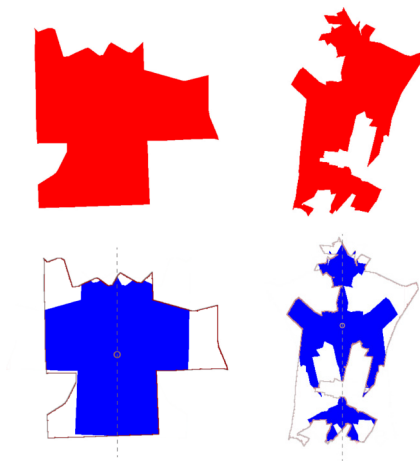
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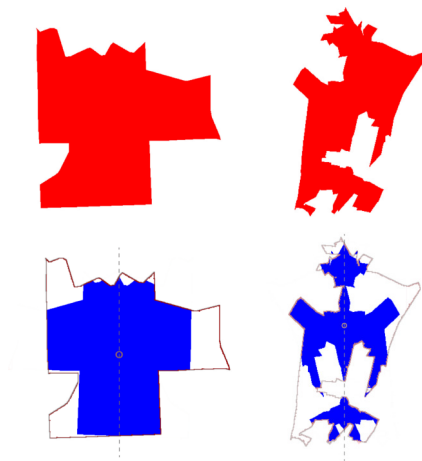
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In both cases, $\text{Overlap} / \text{Original Area} \approx 0.34$

New Measure 2: Number of Visually Significant Corners

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Computer vision algorithm identifies “objects” in photos

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Computer vision algorithm identifies “objects” in photos

~> Fewer corners is more compact

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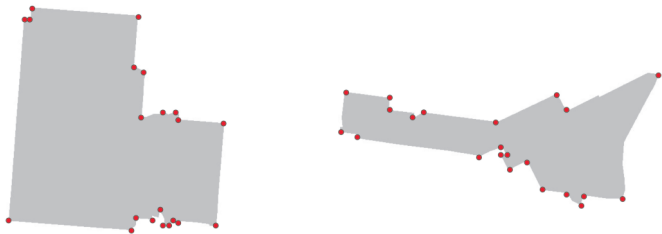
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New Measure 2: Number of Visually Significant Corners

Computer vision algorithm identifies “objects” in photos

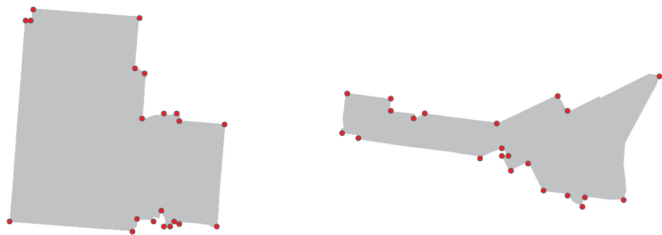
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New Measure 2: Number of Visually Significant Corners

Computer vision algorithm identifies “objects” in photos

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Both districts have 21 significant corners

Which is more compact?



Which is more compact? Depends on the standard!



Which is more compact? Depends on the standard!



Reock

1

2

3

4

Which is more compact? Depends on the standard!



Reock

1

2

3

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Convex Hull

4

3

2

1

Which is more compact? Depends on the standard!



	1	2	3	4
Reock	1	2	3	4
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- **Unusual?** From 18,215 Congressional and State Legislative Districts, we found 162 trillion others (about 0.15%)
- **Many more inconsistencies on individual districts**

Spanning the Academic–Legal Divide

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Utterly fails on inter- and intra-coder reliability

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Full Ranking

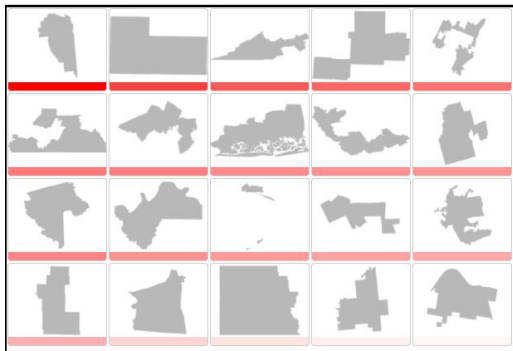


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Full Ranking — on line

MOST Compact Here



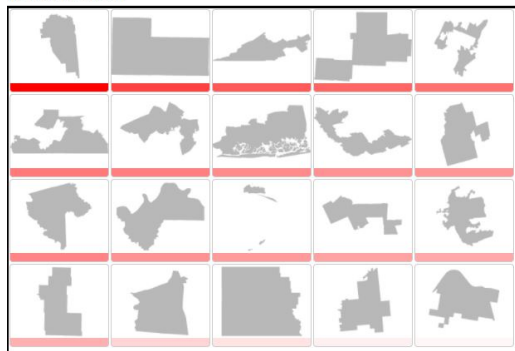
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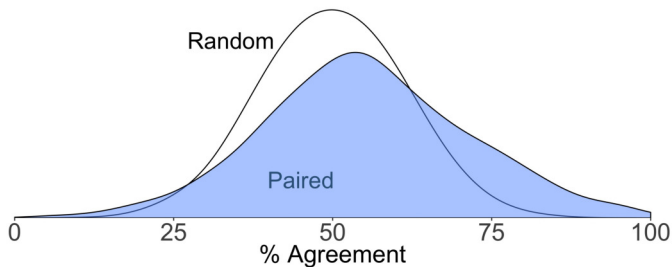
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Ranking: users choose **one dimension** for all evaluations

Intercoder Reliability of Pairs

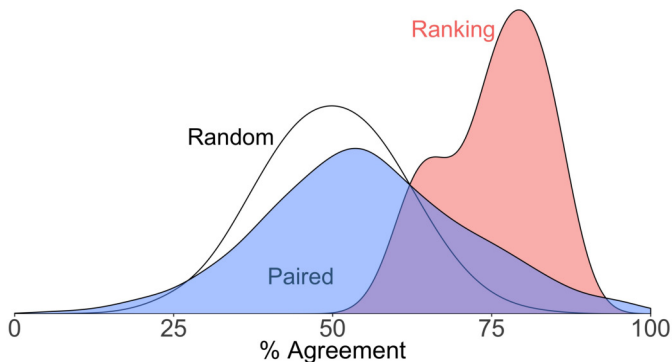
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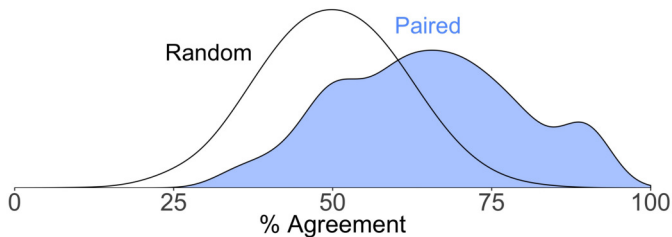
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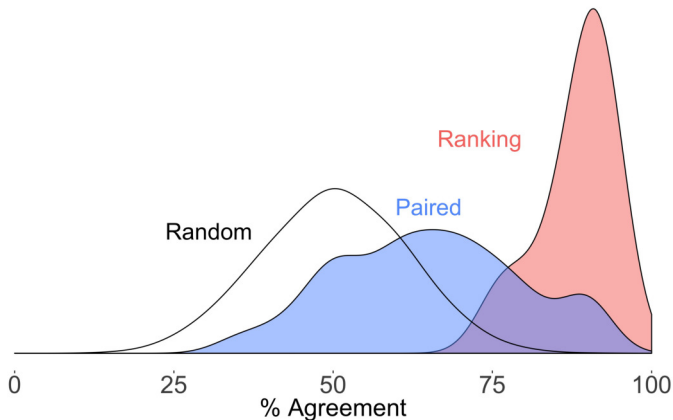
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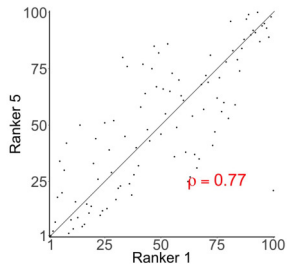


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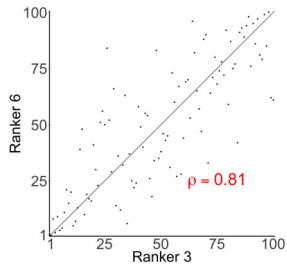
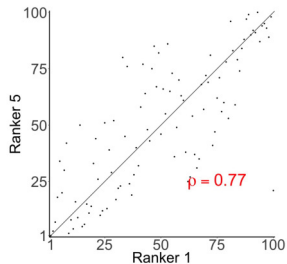
Paired Comparisons: better than chance; Ranking: much better



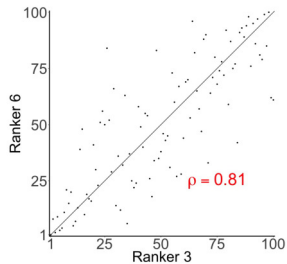
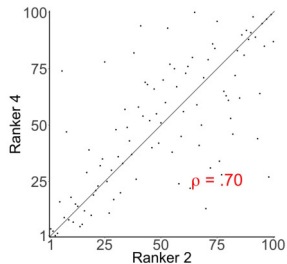
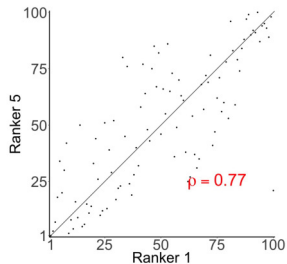
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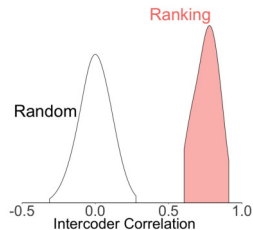
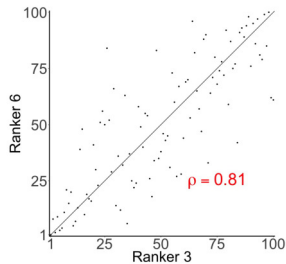
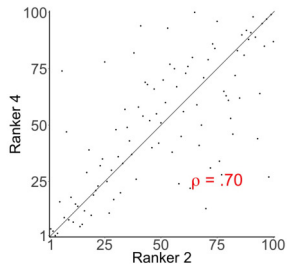
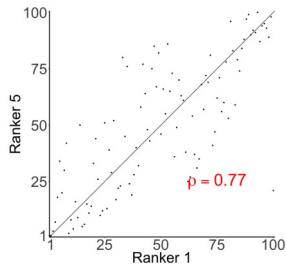
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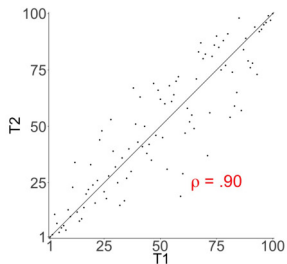
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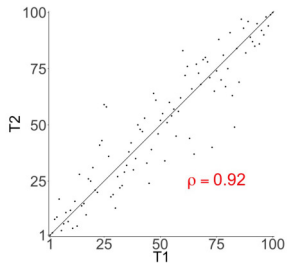
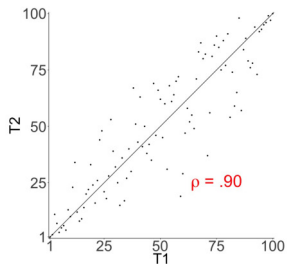
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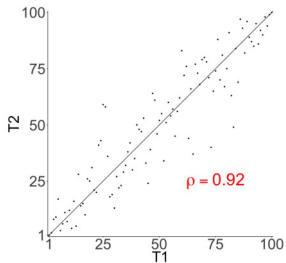
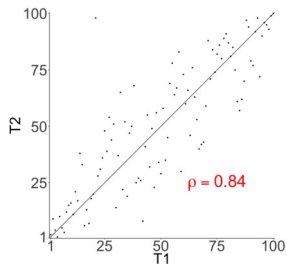
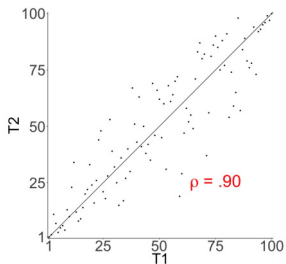
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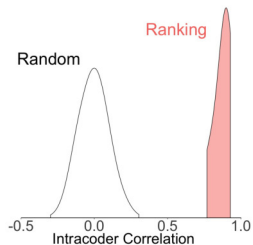
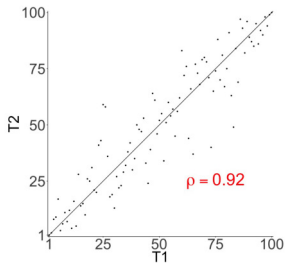
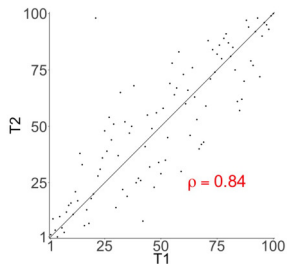
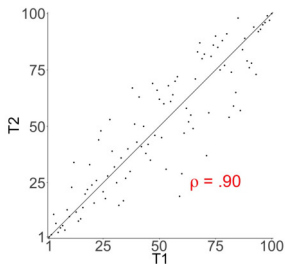
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Goal: Compactness score = $f(\text{shape})$

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- **Ensemble of predictive methods:** least squares, AdaBoosted decision trees, SVM, random forests. . .

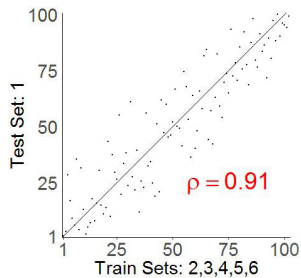
Model Validation: 6-Fold Cross-validation

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Predict Test Set from 5 Training Sets

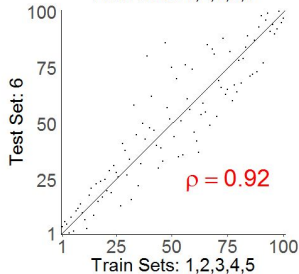
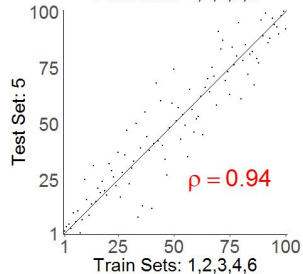
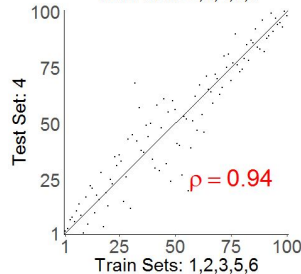
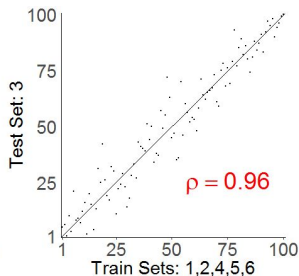
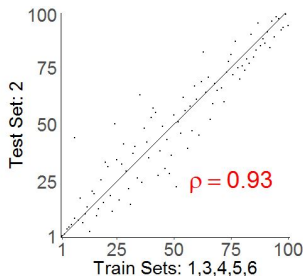
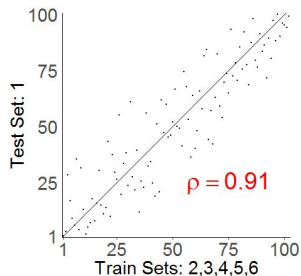
Model Validation: 6-Fold Cross-validation

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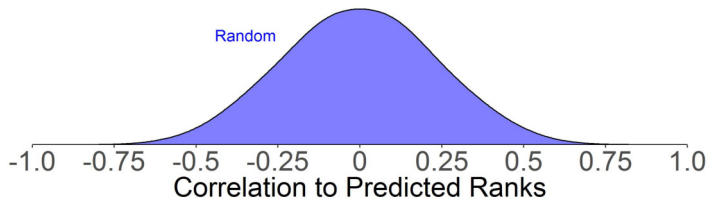
Model Validation: Diverse Respondents

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Respondents ranging from ordinary citizens to those responsible for redistricting

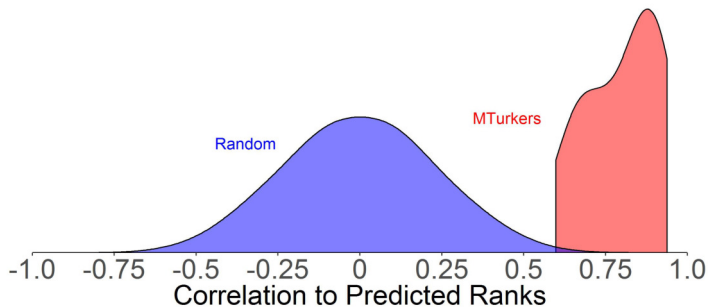
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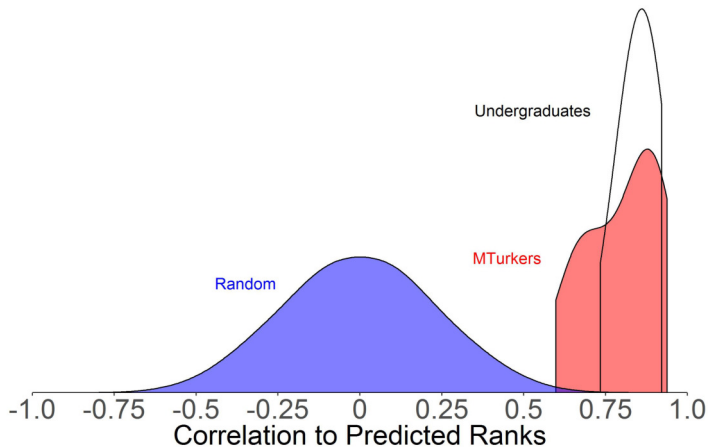
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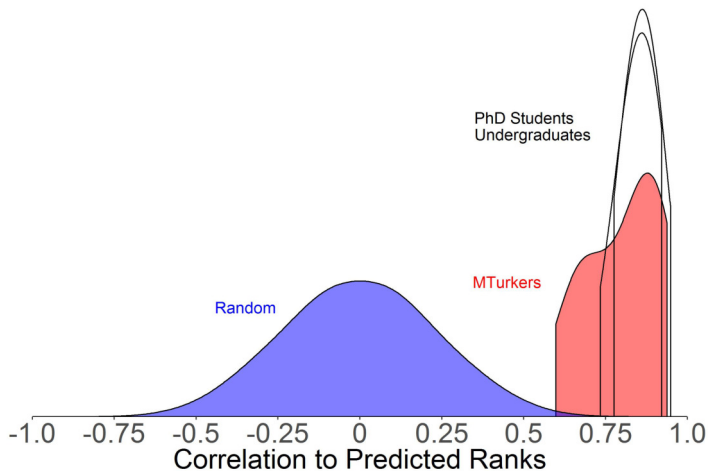
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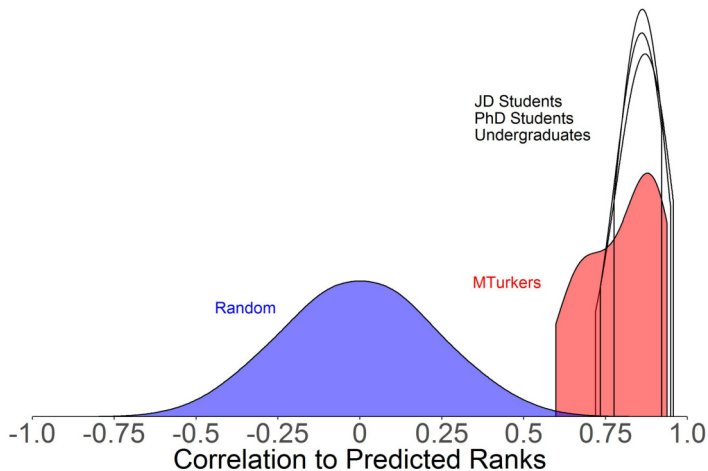
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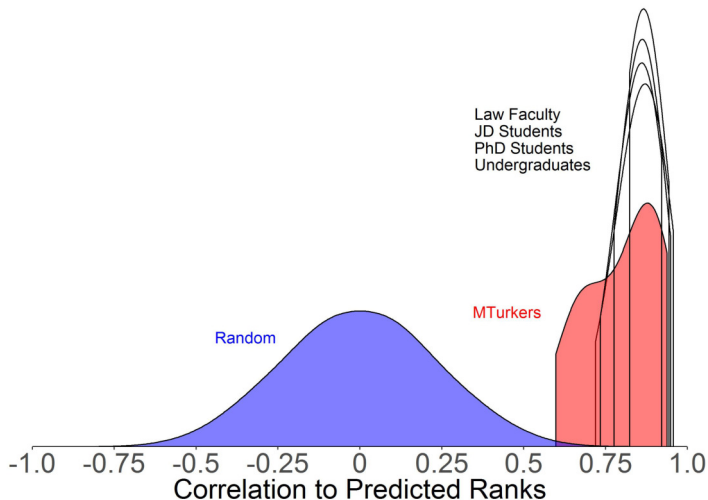
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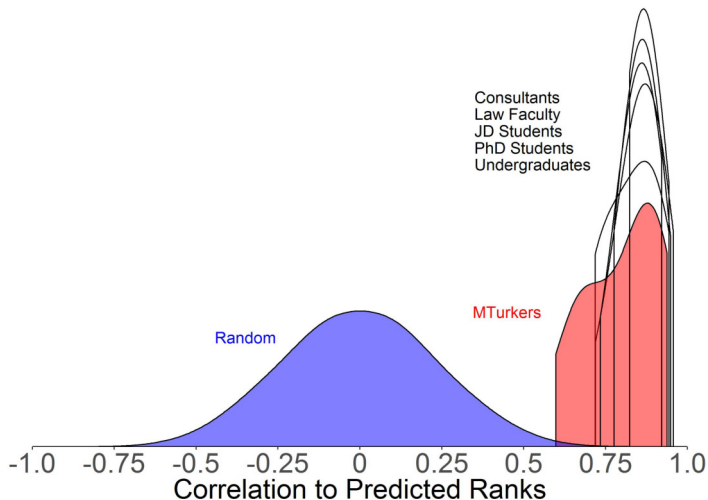
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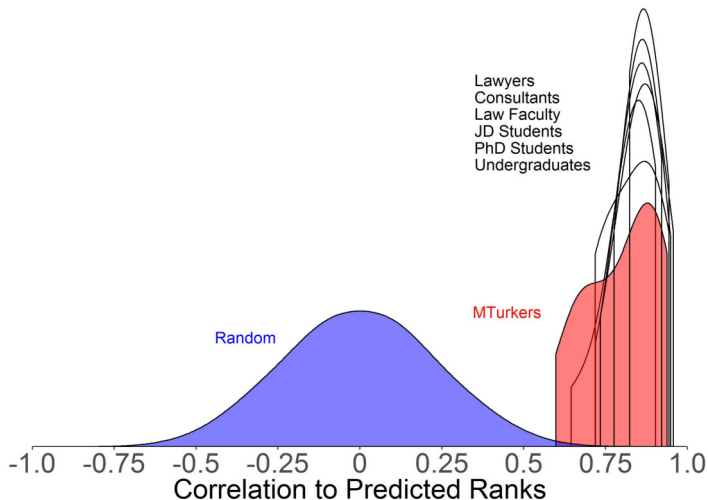
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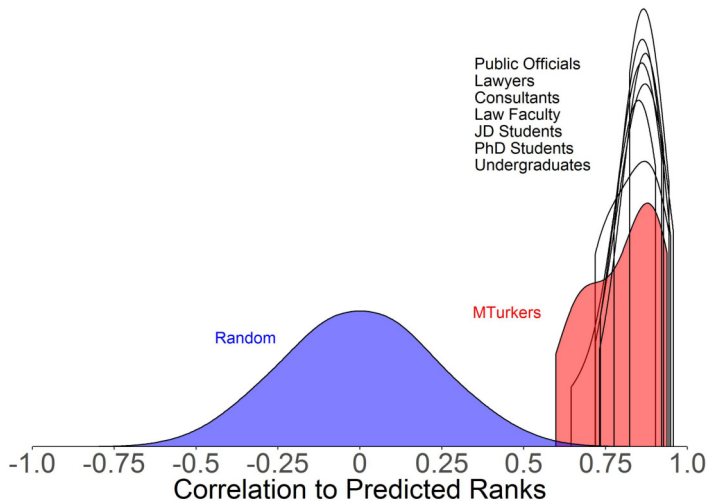
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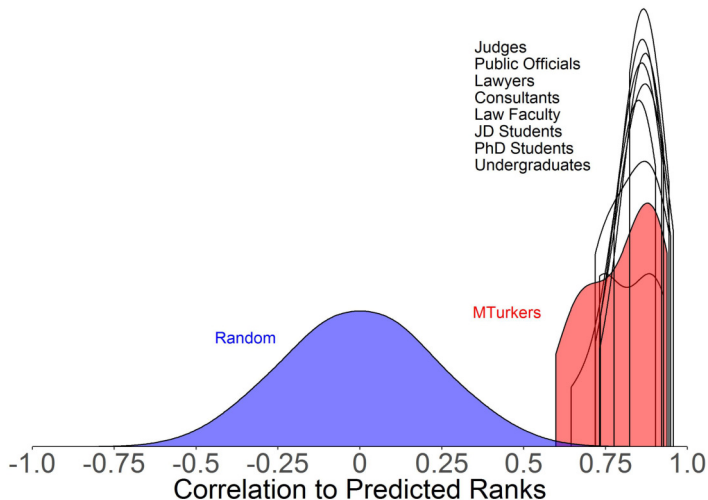
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For more information



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Paper, data, software, slides: j.mp/Compactness